in wind speed, precipitation, humidity, and temperature. These environmental factors can affect aerodynamics, sensor performance, battery life, and communication systems, limiting operational efficiency and mission success. Therefore, developing strategies and technologies to address these environmental constraints is crucial for enhancing drone endurance, resilience, and adaptability across diverse conditions.

Modern research and engineering efforts are increasingly focused on weatherproofing drone designs and integrating adaptive flight control technologies that allow UAVs to function safely in challenging conditions. By improving both hardware and software resilience, drones can expand their range of applications from military reconnaissance and disaster response to agriculture and logistics, regardless of environmental unpredictability.

7.9.1 Drone performance in extreme weather conditions

Weather extremes, such as strong winds, heavy rainfall, extreme temperatures, or dense fog, pose serious operational challenges for drones. High wind speeds can destabilize flight, strain motors, and reduce battery efficiency, often leading to crashes or mission failure. Even advanced stabilization algorithms may struggle to maintain steady positioning under gusty conditions. Similarly, rain and humidity can interfere with electronic components, degrade sensor accuracy, and increase the risk of short circuits. Icing and condensation on propellers or sensors during cold-weather operations can further compromise aerodynamics and visibility.

Temperature variations also significantly affect drone performance. Cold environments reduce battery capacity and chemical reactivity, shortening flight times and limiting power output. Conversely, high temperatures can cause battery overheating, material deformation, and reduced electronic